NOTE Nested `act` claims don't indicate that service77 is pretending to be service16 pretending to be Alice! Think of it as a mask being passed from actor to actor, rather than a single actor wearing multiple layers of masks.

Listing 11.15 An exchanged access token introspection response

```
{
   "aud":"https://service26.example.com",
   "iss":"https://issuer.example.com",
   "exp":1443904100,
   "nbf":1443904000,
   "sub":"alice@example.com",    #A
   "act":    #B
   {
     "sub":"https://service16.example.com",    #B
     "act":    #C
     {
       "sub":"https://service77.example.com"    #C
     }
   }
 }
```

#A The effective user of the token
#B The service that is acting on behalf of the user
#C A previous service that also acted on behalf of the user in the same request

Token exchange introduces an additional network roundtrip to the AS to exchange the access token at each hop of servicing a request. It can therefore be more expensive than the phantom token pattern and introduce additional latency